



Your full name (what appears on your passport): Anandita Rohan Nevrekar

Name of teacher: Maram Haddad

What classes has the teacher taught you? AP Calculus AB

Interested Majors: Engineering, Math, Computer Science

### **What excites you about going to university?**

The thing that excites me the most about university is firstly independence. My whole life I have participated in activities, subjects, and interacted with many different people, after living in three different countries. However I think that when I get to go to university, I will be able to learn how to do the things I love, and manage my life solely by myself teaching me the really important skill of independence. I am also excited about going to university because I think in High School we learn very structure based, and not necessarily every single subject is something that I am passionate about, or something that I love. I think that in University I will have choice and opportunity to make decisions by myself that can help me learn and grow to the best of my abilities. I personally am interested in either a STEM related field like something to do with Math or Computer Science, or something in the engineering field. I think that I see myself following that path because I myself am a very detailed, organised and disciplined person and I feel that creativity is sometimes not my strongest suit. But I am excited for this because I've also noticed that all STEM related fields lack women in them. Hence why I helped lead Girls Can Code (A club promoting coding to elementary girl students). I think that by entering that male dominated field, I would be helping the generations of girls after me to make their mark and be as involved in the up comings of science and math around us.

### **What are you most proud of achieving in the classes you've taken with this teacher?**

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I think my most proud achievements in AP Calculus AB was my mock exam which was the first actual full exam I had taken in calculus that would be the same as what the final AP exam was. I studied really hard in the weeks leading up to the exam, both in class, WIN Sessions, and at home by myself. While I took the exam instead, of feeling stressed and overwhelmed which I had felt quite a few times before, I felt at ease and confident in myself and my abilities. This test was definitely one of my best scoring tests, but more than the actual number score that I got, I was more proud in my ability, to comprehend, and put all that I had learned throughout the whole year into the test. This definitely helped me prove to both myself and my teacher that my skills were transferable and had really developed over the course of the year.

### **What topics were you most interested in this subject and why?**

I think the topic I found the most interesting and that definitely made me question the ideas and overthinking behind calculus was Unit 8. This unit focused on volume, rotations, and calculations that moved beyond algebraic concepts and more into the realm of geometry. Geometry was a subject that I had previously struggled with and never really enjoyed. However, seeing it through the lens of calculus changed my perspective. When I started working with the axis of rotation and calculating volume using integrals, I found it fascinating how these very different ideas could be used to model real world objects and scenarios. It helped me understand that calculus is not just a set of rules for solving equations but a powerful tool for understanding the physical world. Whether it is used to design curved surfaces in architecture, calculate the volume of tanks and containers in engineering, this application of calculus made the subject feel more meaningful and relevant. It gave me a sense of how math connects to real life situations, which made the topic more engaging and enjoyable.

**Were there papers/projects/assignments that you enjoyed the most or found impactful? Why?**

I think that I found the final Water Tank Project the most impactful in calculus, like how I said before about the rotations and volumes, it takes calculus skills and applies them to real life issues and situations to help the greater good. I personally did my project about Uganda because I travelled there in February for a service conference and I was able to learn that Longaroe Sub County in Uganda struggles with water scarcity and lack of clean water. Over 10 million Ugandans face poor access to sanitation, hygiene, and nutrition, a functional water tank could greatly benefit this community, because of low rainfall makes water storage especially important. I was able to apply my calculus skills of the disk method and optimisation to create a water tank that would be most useful and productive towards a family in a village in Uganda. I think a lot of the times in math classes we learn things, like algebra and geometry without knowing how we could ever apply these skills into the real world, and for the first time. I felt like what I was learning in math could actually have a positive impact to the people and places around me and I think that was a really impactful project for me to learn about in calculus.

**Could you share an example of a time in the course when you faced a challenge? How did you overcome it?**

I think the time in the course that I really felt challenged was starting second semester when we were learning integration. I had just become really comfortable with derivation and the applications it had. I saw myself really struggle to just understand what integration was and how it worked. This was also reflected in some of my test scores, because they were dropping. However instead of giving up, and not trying to grow from the mistakes I was making on these new units, I started going to more WIN sessions, staying longer in class, and going over my mistakes that I made on the prior tests. This allowed me to clear up all of the doubts I was having and made me realise that I could

work through all of the misconceptions I was having with integration. I think that for me, overcoming this challenge started with taking accountability and seeking help.

Because for me a lot of the times I try to face challenges by myself, but once I was able to reach out for help from Mrs. Haddad, she personally helped me see through every single issue I was facing and through the extra time I put in, I felt confident in myself and my skills, and was able to do better on the tests after that.

**Could you share an example of a time when you worked collaboratively with others in the class and learned from them? What did you learn/how did it benefit you?**

I think that throughout calculus I've been working very collaboratively with all my peers and classmates. But I think the most impactful collaboration was a few weeks before the actual AP Exam, my peers and I worked together going through every single question on a past MCQ. Through this I was not only able to clear my doubts about questions that I thought were confusing but I was also able to see another approach and view to how I could answer questions. Every single one of my peers had a different way that they looked and approached every single question, and that allowed me to learn, that when I'm stuck on a question, I just need to approach the problem from a different angle. I think that this can obviously be viewed for so many different classes, but I very clearly saw it in calculus, because there's always a right or wrong answer, but there are so many ways that we can reach that answer, and collaboration, communication taught me that.

**What would I like for you to emphasize about me?**

I would really appreciate if you could emphasise how as a student my strengths are my disciplines, resilience and hard work. More importantly how I am someone who give 100% effort to all that I do, and seek as much feedback to improve on the mistakes I



have made. As well as that I use my failures as learning opportunities and don't let them stop me, and continue with determination. I have also used this learning of mistakes to change my mentality of being a student and putting in hard work, and taking initiative and holding myself accountable. Other than just me as a student, I really value and treasure the impact I have on other people hence my involvement with Cancel Period Poverty and Girls Can Code which are clubs that I lead in order to help other girls like myself flourish and grow and have equal opportunities to the people around them. All of these skills would make me hopefully suitable for a STEM related field and, from living in different countries my open mindedness, curiosity, and desire to keep learning from the people and places around me remains strong. As well as that I would hope to use those skills in whatever work, project, college, or place I go to.